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EX NO: 14	**5G Throughput vs SNR with Modulation Labels**

AIM: To simulate the performance of a 5G wireless communication system using MATLAB

by analyzing the relationship between Throughput (Mbps), Signal-to-Noise Ratio (SNR) (dB),

and different Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM).

Procedure:

1. Define the simulation parameters such as SNR range, bandwidth, modulation schemes

(QPSK, 16-QAM, 64-QAM, and 256-QAM), and transmission settings.

2. Generate random binary data and modulate it using the selected modulation schemes.

3. Simulate the transmission of the modulated signals through an Additive White Gaussian

Noise (AWGN) channel with different SNR values.

4. Calculate the Throughput (Mbps) achieved for each modulation scheme at different

SNR levels.

5. Analyze the impact of Signal-to-Noise Ratio (SNR) on communication performance and

throughput.

6. Compare the performance of QPSK, 16-QAM, 64-QAM, and 256-QAM under identical

channel conditions.

7. Plot MATLAB graphs for Throughput vs SNR and SNR vs Modulation Scheme to

visualize the system performance.

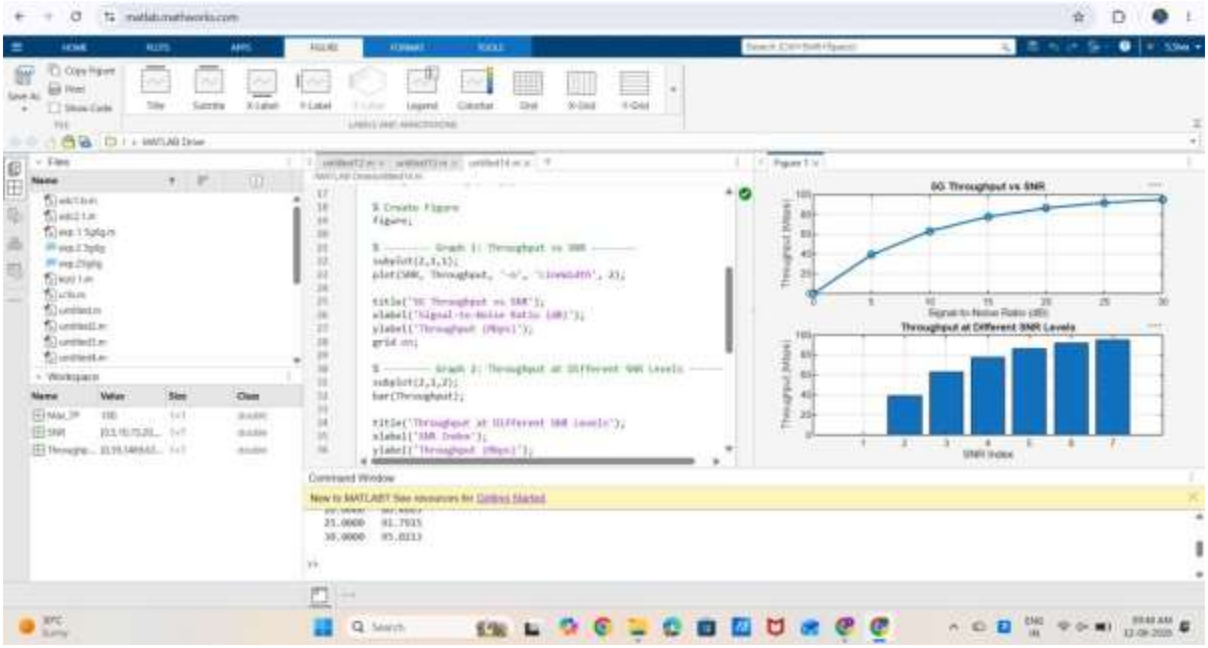
8. Interpret the results to identify the most suitable modulation scheme for different SNR

conditions in a 5G communication system.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ------Graph 1------
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ------Graph 2------
subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index')
ylabel('Throughput (Mbps)')
grid on
```

OUTPUT:



Result:

- The throughput of the 5G system increases as the SNR increases due to improved signal quality.
- Higher SNR enables efficient data transmission, resulting in greater throughput and better communication performance